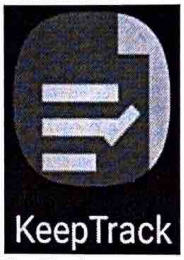
Mobile Application Development – CA1: Review 1

Idea & Proposal Review

Milestone: Problem definition, concept, and project scope

Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Title	KeepTrack – Smart Personal Record and Document Management Application
App Name	KeepTrack
Problem Statement	Users often keep receipts, warranties, insurance details, service records, rental agreements and other important documents in different places such as physical files, email attachments, gallery folders and messaging applications. This makes information difficult to locate quickly and increases the possibility of missing warranty or expiry dates. KeepTrack addresses this problem through one organized, device-based application.
Abstract	KeepTrack is an Android application for storing, organizing and retrieving personal product, warranty, insurance, service, agreement and document information. Users can create category-based records, manage expiry dates, save receipt/document images, search records and use OCR to extract readable text. Local persistence and backup/restore functionality are included to improve reliability and data protection.
Target Users	Students, households and individual users who need a simple way to maintain personal product records, warranties, insurance information, service details, agreements and important documents.
Planned Core Features	1. Category-based record creation and management 2. Warranty and expiry-date tracking with alerts 3. Receipt/document image capture and gallery selection 4. Search, favorites and organized record viewing 5. OCR, backup/restore and import/export
App Icon	 KeepTrack

Remarks:

Name 1 (USN)	Name 2 (USN)	Faculty In-charge
Spencer Aaron Fernandes 25MCAR0123	Mohith T R 25MCAR0227	Mr. Aldrin Jose Antony
Signature:	Signature:	Signature: 01/9/26

Signatures of Members **Signature of Faculty In charge**

Mobile Application Development – CA1: Review 2

UI/UX Design Review

Milestone: Wireframes, mockups, and visual design language

Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Title	KeepTrack – Smart Personal Record and Document Management Application
App Name	KeepTrack
Screens Designed	Onboarding; Profile Setup; Home; Records; Category Records; Add Record; Record Details; Edit Record; Scan Document; Saved Documents; Document/OCR View; Settings; Notifications/Preferences; Backup/Import; About Us.
Wireframe / Mockup Tool	The available project material documents the UI implementation in Android Studio using Jetpack Compose. No separate Figma/Adobe XD file was provided, so no external wireframing tool is claimed.
Colour Palette & Typography	Primary visual language uses KeepTrack blue with light-blue application surfaces, white/light backgrounds, dark readable text and Material 3 typography. A dark-theme option is also provided through the application settings.
Design Consistency Notes	Material 3 components, reusable cards/buttons, consistent spacing, readable typography, category labels, clear primary actions and contrast-aware colours are used throughout the application.
Layout & Visual Balance	Screens use structured vertical layouts, cards, two-column category grids where appropriate, consistent horizontal padding and spacing. The application header respects the system status-bar safe area so the app title does not overlap notification icons or the camera cutout.
Add-ons as per Remarks	Implemented a light-blue app header, dynamic category visibility based on available data, expiry badges that clear after opening a category, saved-document display, light/dark theme support, About Us section and improved safe-area handling.
Evaluation – UI Design & Visual Appeal	Designed for a clean, practical document-management experience with consistent Material 3 components and clear navigation.

Remarks:

Name 1 (USN)	Name 2 (USN)	Faculty In-charge
Spencer Aaron Fernandes 25MCAR0123 Signature:	Mohith T R 25MCAR0227 Signature:	Mr. Aldrin Jose Antony Signature: 18/4/26

Signatures of Members **Signature of Faculty In charge**

Mobile Application Development – CA1: Review 3

Development & Responsiveness Review

Milestone: Core build progress and responsive layout implementation

oup Number	21
m Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
e	KeepTrack – Smart Personal Record and Document Management Application
Name	KeepTrack
tures Implemented	Record creation, editing, viewing and deletion; category-based records; search; favorites; warranty calculation; expiry indicators/badges; notifications settings; camera capture; gallery selection; persistent saved documents; OCR/text extraction; local Room persistence; JSON import/export; backup/restore; Google Drive file-picker workflow; delete-all records; profile/settings; About Us; light/dark theme; bottom navigation.
h Stack Used	Android platform; Kotlin 2.2.10; Jetpack Compose / Material 3; Room 2.8.3; ViewModel + Repository architecture; Jetpack Navigation Compose 2.9.6; Android MediaStore; Google ML Kit Text Recognition; JSON import/export; Gradle Version Catalog.
ices / Screen Sizes Tested	Physical Samsung Android smartphone (testing shown in the supplied video/screenshots) and Google Pixel 8 emulator. The app is primarily portrait-oriented.
ponsive Design Approach	Jetpack Compose layouts use fillMaxWidth/fillMaxSize, LazyColumn, LazyVerticalGrid, weight-based rows/columns, adaptive spacing and system-bar padding. Category cards use a two-column grid, while content remains vertically scrollable on smaller displays.
wn Issues / Bugs	Final verification is still required for complete end-to-end camera/gallery save-and-reopen behaviour, backup/restore edge cases and final build verification. Earlier navigation/URI encoding and theme-setter compilation issues were identified during development and addressed in subsequent code revisions.
ons as per Remarks	Added permanent MediaStore image storage, saved-document thumbnails, safe URI navigation, status-bar-aware header, dynamic categories, expiry badge clearing, dark-theme preference persistence and improved settings/about content.
ation – Responsiveness & Interactivity	Core screens and interactions are implemented with Compose-based layouts designed for portrait Android devices.

ks:

1 (USN)	Name 2 (USN)	Faculty In-charge
er Aaron Fernandes AR0123	Mohith T R 25MCAR0227	Mr. Aldrin Jose Antony
ure:	Signature:	Signature: 13/2/26

Signatures of Members **Signature of Faculty In charge**

Mobile Application Development – CA1: Review 4

UX, Navigation & Usability Testing

Milestone: User flow, interactivity, and usability feedback

Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Title	KeepTrack – Smart Personal Record and Document Management Application
App Name	KeepTrack
Navigation Flow Summary	The application uses bottom navigation for Home, Records, Scan and Settings. From Home, users can open categories, view expiry information, add records and scan documents. Records lead to category lists and details, while Scan leads to camera/gallery selection, saved documents and OCR/document viewing. Settings provides profile, notification, theme, backup/import and data-management options.
Interactive Components Used	Buttons, Material 3 Cards, NavigationBar/NavigationBarItem, text fields, search bar, category cards, expiry badges, favorites, date pickers/calendars, switches, dialogs, image/document pickers, camera capture, lists/grids, delete confirmation, backup/import file pickers and selectable OCR text.
Usability Testing Method	Functional walkthrough testing was performed on an Android device, supported by the supplied testing video. The team tested navigation, record creation, category access, scanning/camera and gallery flows, saved-document visibility, settings and visual behaviour.
Usability Feedback Received	Testing highlighted the need for clearer separation between the application header and system status area, dynamic category presentation, expiry badges that behave like familiar messaging badges, visible saved documents after capture, an accessible theme switch and a clearer About Us section. These changes were incorporated into the UI.
Navigation Improvements Planned	Continue final end-to-end verification of camera/gallery navigation, confirm saved-image persistence after restart, verify backup/restore, and perform a final pass for accessibility, spacing and device-specific behaviour.
Add-ons as per Remarks	Introduced a status-bar-safe light-blue header, dynamic categories, WhatsApp-style expiry badge clearing after category opening, persistent saved-document list, theme toggle with saved preference, About Us section and clearer Settings organization.
Evaluation – User Experience & Navigation	The navigation is organized around the four primary tasks—Home, Records, Scan and Settings—with supporting category and detail flows.

Remarks:

Name 1 (USN)	Name 2 (USN)	Faculty In-charge
Spencer Aaron Fernandes 25MCAR0123 Signature:	Mohith T R 25MCAR0227 Signature:	Mr. Aldrin Jose Antony Signature: 18/9/26

Signatures of Members **Signature of Faculty In charge**

Mobile Application Development – CA1: Review 5

Final Presentation & Deployment Review

Milestone: Completed app, deployment, and final demo

Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Project Title	KeepTrack – Smart Personal Record and Document Management Application
App Name	KeepTrack
Deployment Details	Android application developed and tested through Android Studio on Android hardware/emulator. The available project evidence does not indicate a public Play Store release or public hosting URL; deployment is therefore documented as local APK/device testing.
Final Feature Summary	KeepTrack provides category-based personal record management; add/edit/delete/view; search and favorites; warranty and expiry calculation; expiry badges/alerts; camera and gallery document capture; persistent MediaStore saved documents; OCR text extraction/copy; Room local storage; JSON import/export; backup/restore; Google Drive file-picker backup workflow; profile/settings; notification settings; light/dark theme; bottom navigation; responsive portrait UI; About Us.
Demo Notes	Live demo should show: launch and header/status-bar separation; Home and dynamic categories; opening a category and clearing its expiry badge; Add Record with category-specific fields; record details/edit/delete; search/favorites; Scan → Take Photo or Gallery → Saved Documents; document/OCR viewing; Settings → theme switch; backup/import and delete-all confirmation.
Project Report Submitted	Yes – the CA1 project review document is being submitted with this completed Review 1–5 content.
App Icon	 KeepTrack
Add-ons as per Remarks	Final UI refinements include the safe-area header, dynamic category display, expiry-badge clearing behaviour, persistent document storage/viewing, light/dark theme preference, About Us content, and improved settings organization. Final verification of all device-specific edge cases remains part of the demonstration/testing process.
Final Evaluation – CA1 Rubric Total 15 Marks	Faculty evaluation / marks: _____ / 15

Remarks:

Name 1 (USN)	Name 2 (USN)	Faculty In-charge
Spencer Aaron Fernandes 25MCAR0123	Mohith T R 25MCAR0227	Mr. Aldrin Jose Antony
Signature: _____	Signature: _____	Signature: _____ 18/9/26

Signatures of Members **Signature of Faculty In charge**

Mobile Application Development – CA3: Review 1

Idea & AI Use-Case Proposal

Milestone: Problem definition, AI/ML use case, and project scope

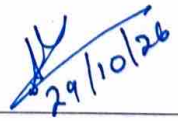
Section	Details
Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Title	AI-Based Receipt Scanning and Warranty Management in KeepTrack
App Name	KeepTrack
Proposed AI/ML Feature	Vision AI extracts structured receipt/document details: store, product, category, brand, model, serial number, purchase date/price, warranty period and notes. Results are editable before saving.
Why AI is Needed	Manual entry is slow and error-prone. AI automates initial extraction from camera/gallery images and organizes data into fields. Users verify/correct results; AI output is not guaranteed accurate.
Target Users	Consumers managing receipts, electronics and household-appliance purchases, product details, purchase history and warranty information.
Add-ons as per Remarks	Android app combines camera/gallery selection, AI extraction, editable review and Room storage. Login, registration, persistent sessions and logout are implemented. Records are saved only after user confirmation.

Remarks:



Spencer Aaron Fernandes (25MCAR0123)
Mohith T R (25MCAR0227)

Signature of Members



Signature of Faculty In Charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 2

AI/ML Model & Data Review

Milestone: Dataset preparation, model selection, and training approach

Section	Details
Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123) Mohith T R (25MCAR0227)
Title	AI-Based Receipt Scanning and Warranty Management in KeepTrack
App Name	KeepTrack
Dataset Used	Real receipt/document images captured by camera or selected from gallery were used for testing. No separate labelled dataset, image count, annotations or train/test split has been reported. A pre-trained API model is used; no custom training is reported.
Data Preprocessing	Original saved via FileStorage; resize maximum dimension to 1024 px; convert to JPEG; compress at 80% quality; Base64 encode as an image data URL; send with extraction instructions; parse response into structured editable fields.
Model / Algorithm Chosen	Pre-trained multimodal vision model `qwen/qwen3.8-27b` via Groq API, chosen to interpret document images and extract purchase/product fields. Endpoint: https://api.groq.com/openai/v1/chat/completions. Retrofit, OkHttp and Kotlin serialization support integration.
Training & Validation Approach	Validation focuses on unit tests, API request/response handling, live image trials and parsing into review fields. No custom training or train/test split reported; these checks do not establish formal model accuracy.
Model Performance (so far)	Reported agent test: 9 unit tests passed, 0 skipped, 0 failed; one live API request returned HTTP 200 OK in 784 ms; a non-receipt image was identified as an ID card. No accuracy/precision/recall/F1 score reported; timing applies to that request.
Integration Plan	Image → local original-image storage → preprocessing → Groq request → structured response parsing → editable review → user correction → Confirm & Save → Room. Cancelling creates no Room record; original image remains stored. Production should protect API credentials with a secure backend proxy.
Add-ons as per Remarks	Image preprocessing, Groq integration, structured extraction, editable fields, retry/manual-entry path and user-controlled saving are implemented/reported. Formal dataset-based evaluation is unavailable.

Evaluation – AI Integration & Performance (partial)

Remarks:

Spencer Aaron Fernandes (25MCAR0123)
Mohith T R (25MCAR0227)

Signature of Members

29/10/25

Signature of Faculty In Charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 3

App Development & Functionality Review

Milestone: Core app build and functional integration of the AI feature

Section	Details
Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123); Mohith T R (25MCAR0227)
Title	AI-Based Receipt Scanning and Warranty Management in KeepTrack
App Name	KeepTrack
Features Implemented	Login/Register, Home, Records, Scan Document, image preview/loading, AI extraction, editable review, Confirm & Save, Settings and logout. Camera/gallery input, local image storage and Room persistence support the workflow.
Tech Stack Used	Android Studio; Kotlin; Jetpack Compose; Room; Retrofit; OkHttp; Kotlin serialization; Groq API; `qwen/qwen3.8-27b`; FileStorage. Google ML Kit OCR is included in architecture notes; verify its exact runtime role against source code.
AI Feature Functionality	Capture/select image; save original locally; resize to max 1024 px, JPEG at 80%, Base64 encode; call Groq model; parse seller/product/category/brand/model/serial/date/price/warranty/notes; display editable fields; save to Room only after Confirm & Save.
Error Handling / Edge Cases	Preview/loading state during extraction; error with retry/manual-entry path; editable values for corrections; cancellation before confirmation creates no record; original image stored separately. Early-form-display and Back-navigation bugs were reported fixed and tested. Exhaustive testing of every failure case is not claimed.
Known Issues / Bugs	Two previously reported bugs—editable fields appearing before extraction completed and Back-navigation glitches after gallery selection—were reported fixed and tested. No other confirmed unresolved issue was supplied; this does not prove no other bugs exist.
Add-ons as per Remarks	Login, registration, persistent session and logout implemented. Valid returning session opens Home; logout in Settings ends session and returns to Login while preserving Room records and stored images. AI output requires user review before saving.

Evaluation – Application Functionality

Remarks:

Spencer Aaron Fernandes (25MCAR0123)
Mohith T R (25MCAR0227)

Signature of Members

29/10/25

Signature of Faculty In Charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 4

UI/UX & AI Output Presentation Review

Milestone: Interface design and clarity of AI-generated results

Section	Details
Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123); Mohith T R (25MCAR0227)
Title	AI-Based Receipt Scanning and Warranty Management in KeepTrack
App Name	KeepTrack
Screens Designed	Login, Registration, Home, Records, Scan Document, document preview/loading and editable review, and Settings. Verify exact final screen set against the current build.
Navigation Flow	Authentication: first-time user → Login/Register → Home; valid returning session → Home; Settings → Logout → Login. Scan: camera/gallery → save original → preview/loading → AI extraction → editable review → Confirm & Save → Room. Cancellation before confirmation creates no record and does not delete the original image.
AI Output Presentation	Structured editable fields: store, product, category, brand, model, serial number, purchase date, purchase price, warranty period in months and notes. Processing is separated from review. No formal confidence-score display or measured extraction-accuracy indicator has been documented.
Interactive Components Used	Login/register forms; camera/gallery controls; preview/loading; editable fields; retry/manual-entry; Confirm & Save; record list/navigation; Settings/logout; validation/error messages and logout confirmation where implemented.
Responsiveness Check	Tested on a Samsung Android phone. Exact model, Android version, display dimensions and multi-device results were not supplied. Verify text/buttons fit, keyboard does not obscure fields, long values display correctly and navigation works on each supported size before claiming broad responsiveness.
Add-ons as per Remarks	Editable AI results, preview/loading, confirmation-before-save, retry/manual-entry, persistent sessions and logout improve usability. User verifies AI results; local records and images are intended to remain after logout. Navigation issue was reported fixed and tested.

Evaluation – UI/UX Design

Remarks:

Spencer Aaron Fernandes (25MCAR0123)
Mohith T R (25MCAR0227)

Signature of Members

29/10/26

Signature of Faculty In Charge
Mr. Aldrin Jose Antony

Mobile Application Development – CA3: Review 5

Final Deployment & Demo Review

Milestone: Deployed application, final AI performance, and live demo

Section	Details
Group Number	21
Team Members	Spencer Aaron Fernandes (25MCAR0123); Mohith T R (25MCAR0227)
Title	AI-Based Receipt Scanning and Warranty Management in KeepTrack
App Name	KeepTrack
Deployment Details	Android debug build developed in Android Studio using Kotlin/Jetpack Compose; tested on a Samsung Android phone. Groq endpoint: https://api.groq.com/openai/v1/chat/completions. Public Play Store release, APK link, exact device model and Android version were not provided.
Final AI Model Performance	Model `qwen/qwen3.8-27b` via Groq. Reported: 9 unit tests passed, 0 skipped, 0 failed; one live request HTTP 200 OK; response time 784 ms; sample non-receipt image identified as an ID card. No labelled benchmark dataset or accuracy/precision/recall/F1 score reported.
Final Feature Summary	Login/register and persistent sessions; Home/Records/Settings; camera/gallery input; local original-image storage; preprocessing; Groq extraction; editable product/purchase/warranty fields; retry/manual entry; Confirm & Save; Room persistence; logout while preserving saved local data.
Demo Notes	Demo: login and returning-session flow; logout and verify records remain; select/capture receipt; show preview/loading; display/edit extracted fields; Confirm & Save; open saved record; cancel another scan to show no record is created; demonstrate Back navigation and retry/manual entry if available. Use test data; never expose credentials/private documents.
Project Report Submitted	Complete according to actual submission status. Add the final report attachment/link if required.
Add-ons as per Remarks	Demonstrate authentication, scanning, AI extraction, user verification, local saving and navigation. Early form-display and gallery Back-navigation bugs were reported fixed and tested. Faculty marks/remarks/signatures remain for completion; report only tests and deployment evidence actually available.

Final Evaluation – CA3 Rubric (Total 15 Marks)

Remarks:

Spencer Aaron Fernandes (25MCAR0123)
Mohith T R (25MCAR0227)

Signature of Members

29/10/25

Signature of Faculty In Charge
Mr. Aldrin Jose Antony